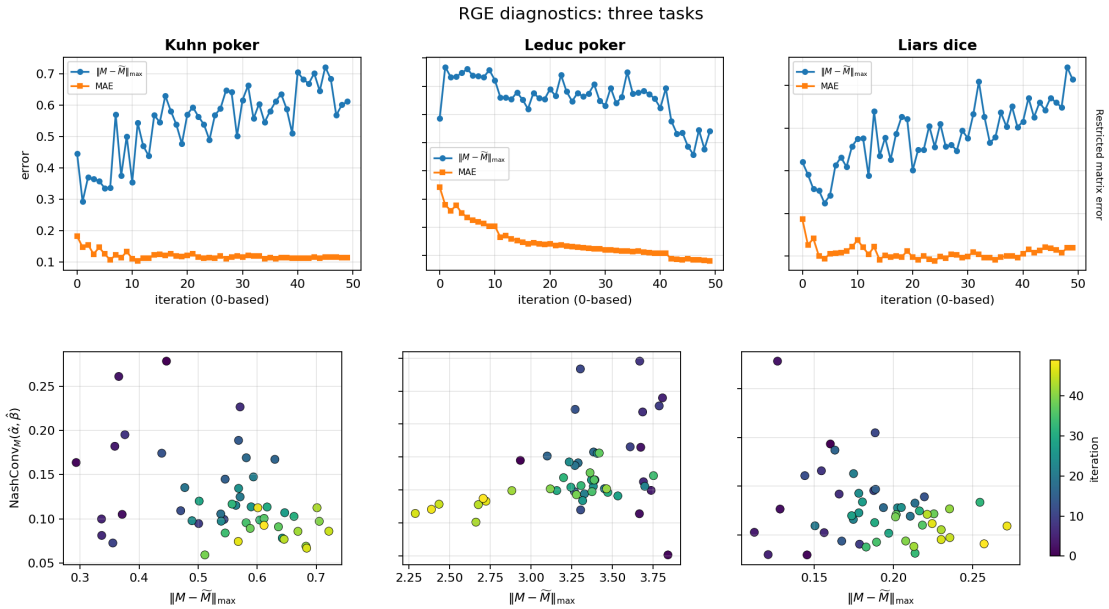

Figure 1. Diagnostics of approximate restricted-game equivalence on three tasks. Columns correspond to Kuhn poker, Leduc poker, and Liar's dice. In the first row, we plot the restricted-game payoff-matrix discrepancy between the true matrix M and the synthesized matrix $\widetilde{M}$ over outer iterations, including $|M - \widetilde{M}|_{\max}$ and MAE. In the second row, we plot the true restricted-game $\mathrm{NashConv}_M(\hat{\alpha}, \hat{\beta})$ against $|M - \widetilde{M}|_{\max}$, where $(\hat{\alpha}, \hat{\beta})$ is computed from $\widetilde{M}$. The figure provides an empirical diagnostic of whether the learned abstraction preserves restricted-game payoffs and equilibrium quality during training. It is intended as an operational proxy for approximate row-space coverage in the learned setting, rather than a direct certificate of the global condition $\mathrm{Row}(P_\nu) \supseteq \mathcal{F}$.